

# E-Commerce Platform Sales Analysis

## Project Objective:

The main objective of my project is to analyze the e-commerce sales database

## It Focus on:

Analyzing sales, product, store, customers database.

cleaning and transformation the raw dataset's .

Combination of data's from the different table's.

With the help of VLOOKUP formulas.

Performing Descriptive Statistical analysis.

Creating different type of pivot table's and chart's.

## Dataset Description:

The project using a e-commerce sales dataset containing the information about sales amount, products, stores and customers.

## Column Description:

| Product table                                                                                                                                         | Description                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Product_id</li><li>• Product_name</li><li>• Category</li><li>• Brand</li><li>• Cost</li><li>• Stock</li></ul> | <ul style="list-style-type: none"><li>• Unique id of the product</li><li>• Name of the product</li><li>• Product category</li><li>• Brand of the product</li><li>• Cost of the product</li><li>• Available product</li></ul> |
| Store table                                                                                                                                           | Description                                                                                                                                                                                                                  |

|                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Store_id</li> <li>• Store_name</li> <li>• Region</li> <li>• City</li> <li>• Store_type</li> </ul>                                                                                                                       | <ul style="list-style-type: none"> <li>• Unique id of the store</li> <li>• Name of the store</li> <li>• Region where the store operates</li> <li>• City of the store</li> <li>• Such as Flagship,online,outlet</li> </ul>                                                                                               |
| <b>Customer table</b>                                                                                                                                                                                                                                            | <b>Description</b>                                                                                                                                                                                                                                                                                                      |
| <ul style="list-style-type: none"> <li>• Customer_id</li> <li>• Name</li> <li>• Age</li> <li>• Gender</li> <li>• City</li> <li>• State</li> <li>• Country</li> <li>• Loyalty_level</li> </ul>                                                                    | <ul style="list-style-type: none"> <li>• Unique customer id</li> <li>• Customer name</li> <li>• Customer age</li> <li>• Gender of the customer</li> <li>• Customer's city</li> <li>• Customers state</li> <li>• Customers country</li> <li>• Customers Loyalty level</li> </ul>                                         |
| <b>Sales table</b>                                                                                                                                                                                                                                               | <b>Description</b>                                                                                                                                                                                                                                                                                                      |
| <ul style="list-style-type: none"> <li>• Sales_id</li> <li>• Order_id</li> <li>• Customers_id</li> <li>• Product_id</li> <li>• Store_id</li> <li>• Quantity</li> <li>• Unit_price</li> <li>• Discount</li> <li>• Payment_type</li> <li>• Total_amount</li> </ul> | <ul style="list-style-type: none"> <li>• Unique sales transaction id</li> <li>• Date of the order</li> <li>• Customer id</li> <li>• Product id</li> <li>• Store id</li> <li>• Quantity available</li> <li>• Price per unit</li> <li>• Discount applied</li> <li>• Way of the payment</li> <li>• Final amount</li> </ul> |

Data cleaning & transformation:

Formula used:

**Customer name cleaning**

=TRIM(CLEAN(PROPER(B2)))

## Countblank:

=COUNTBLANK(A2:G101)

**Median:**

=MEDIAN(G2:G101)

**Average:**

=AVERAGE(F2:F101)

## VLOOKUP:

=VLOOKUP(A2,Table4,3,FALSE)

## Descriptive Statistic

[illegible]

|    | A        | B          | C           | D          | E        | F        | G          | H        | I            | J            | K | L | M | N | O | P | Q | R | S | T | U | V | W | X |
|----|----------|------------|-------------|------------|----------|----------|------------|----------|--------------|--------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1  | Sales_ID | Order_Date | Customer_ID | Product_ID | Store_ID | Quantity | Unit_Price | Discount | Payment_Type | Total_Amount |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 2  | SALE0    | 2024-07-07 | CUST1409    | PROD97     | STORE3   | 1        | ₹ 218.04   | 9%       | Credit Card  | ₹198.42      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 3  | SALE1    | 2023-12-25 | CUST1467    | PROD74     | STORE17  | 3        | ₹ 104.47   | 27%      | PayPal       | ₹228.79      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 4  | SALE10   | 2024-02-13 | CUST1065    | PROD24     | STORE14  | 1        | ₹ 278.63   | 7%       | Credit Card  | ₹259.13      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 5  | SALE100  | 2025-01-10 | CUST1369    | PROD33     | STORE5   | 2        | ₹ 401.50   | 18%      | Credit Card  | ₹658.46      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 6  | SALE1000 | 2024-06-13 | CUST1190    | PROD59     | STORE2   | 2        | ₹ 909.51   | 27%      | Credit Card  | ₹1,327.88    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 7  | SALE1001 | 2025-04-29 | CUST1052    | PROD69     | STORE14  | 4        | ₹ 593.84   | 24%      | PayPal       | ₹1,805.27    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 8  | SALE1002 | 2024-10-01 | CUST1109    | PROD73     | STORE10  | 3        | ₹ 909.11   | 23%      | COO          | ₹2,100.04    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 9  | SALE1003 | 2024-09-07 | CUST1399    | PROD24     | STORE1   | 4        | ₹ 996.85   | 24%      | PayPal       | ₹3,030.42    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 10 | SALE1004 | 2025-06-26 | CUST1300    | PROD00     | STORE17  | 1        | ₹ 522.69   | 6%       | COO          | ₹491.33      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 11 | SALE1005 | 2024-04-23 | CUST1194    | PROD54     | STORE3   | 3        | ₹ 261.05   | 12%      | PayPal       | ₹689.17      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 12 | SALE1006 | 2025-09-18 | CUST1087    | PROD23     | STORE17  | 3        | ₹ 644.73   | 1%       | COO          | ₹1,914.85    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 13 | SALE1007 | 2023-10-18 | CUST1198    | PROD75     | STORE2   | 2        | ₹ 541.52   | 26%      | PayPal       | ₹801.45      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 14 | SALE1008 | 2023-11-29 | CUST1426    | PROD63     | STORE10  | 1        | ₹ 395.18   | 9%       | PayPal       | ₹359.61      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 15 | SALE1009 | 2025-03-24 | CUST1100    | PROD08     | STORE8   | 4        | ₹ 56.37    | 17%      | Credit Card  | ₹187.15      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 16 | SALE101  | 2024-06-10 | CUST1084    | PROD25     | STORE13  | 3        | ₹ 359.56   | 17%      | PayPal       | ₹895.30      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 17 | SALE1010 | 2025-09-08 | CUST1209    | PROD79     | STORE9   | 4        | ₹ 202.59   | 6%       | COO          | ₹761.74      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 18 | SALE1011 | 2024-11-05 | CUST1004    | PROD81     | STORE5   | 2        | ₹ 358.36   | 5%       | PayPal       | ₹680.88      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 19 | SALE1012 | 2024-08-31 | CUST1072    | PROD37     | STORE18  | 4        | ₹ 85.62    | 28%      | COO          | ₹246.59      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 20 | SALE1013 | 2024-06-16 | CUST1185    | PROD07     | STORE17  | 4        | ₹ 939.26   | 23%      | PayPal       | ₹2,862.12    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 21 | SALE1014 | 2025-01-24 | CUST1309    | PROD60     | STORE13  | 2        | ₹ 92.64    | 14%      | COO          | ₹159.34      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 22 | SALE1015 | 2025-03-09 | CUST1248    | PROD62     | STORE4   | 4        | ₹ 139.05   | 29%      | COO          | ₹394.90      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 23 | SALE1016 | 2024-06-18 | CUST1140    | PROD53     | STORE4   | 3        | ₹ 155.87   | 3%       | COO          | ₹453.58      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 24 | SALE1017 | 2025-06-29 | CUST1442    | PROD34     | STORE16  | 4        | ₹ 115.42   | 3%       | COO          | ₹447.83      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 25 | SALE1018 | 2025-06-15 | CUST1122    | PROD52     | STORE6   | 3        | ₹ 593.62   | 8%       | Credit Card  | ₹1,638.39    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 26 | SALE1019 | 2025-03-17 | CUST1353    | PROD45     | STORE4   | 2        | ₹ 825.04   | 20%      | Credit Card  | ₹1,320.06    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 27 | SALE102  | 2024-09-17 | CUST1097    | PROD80     | STORE12  | 2        | ₹ 471.88   | 24%      | PayPal       | ₹717.26      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 28 | SALE1020 | 2024-01-31 | CUST1464    | PROD25     | STORE13  | 1        | ₹ 963.76   | 14%      | PayPal       | ₹828.83      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 29 | SALE1021 | 2024-05-03 | CUST1292    | PROD55     | STORE6   | 1        | ₹ 238.24   | 12%      | PayPal       | ₹209.65      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 30 | SALE1022 | 2024-12-18 | CUST1457    | PROD82     | STORE16  | 1        | ₹ 756.63   | 4%       | PayPal       | ₹726.36      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 31 | SALE1023 | 2025-07-07 | CUST1019    | PROD93     | STORE7   | 3        | ₹ 159.76   | 22%      | Credit Card  | ₹373.84      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 32 | SALE1024 | 2024-03-03 | CUST1368    | PROD95     | STORE8   | 1        | ₹ 329.94   | 14%      | COO          | ₹283.75      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 33 | SALE1025 | 2025-03-04 | CUST1060    | PROD17     | STORE8   | 1        | ₹ 234.06   | 11%      | PayPal       | ₹208.31      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 34 | SALE1026 | 2023-12-18 | CUST1317    | PROD81     | STORE0   | 2        | ₹ 773.75   | 16%      | PayPal       | ₹1,299.90    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 35 | SALE1027 | 2024-05-10 | CUST1436    | PROD52     | STORE18  | 1        | ₹ 73.88    | 4%       | Credit Card  | ₹70.92       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 36 | SALE1028 | 2024-02-15 | CUST1005    | PROD86     | STORE18  | 1        | ₹ 859.30   | 16%      | PayPal       | ₹721.81      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 37 | SALE1029 | 2024-09-29 | CUST1129    | PROD81     | STORE6   | 4        | ₹ 762.43   | 5%       | Credit Card  | ₹2,897.23    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 38 | SALE103  | 2025-09-07 | CUST1182    | PROD34     | STORE4   | 4        | ₹ 925.19   | 19%      | PayPal       | ₹2,997.62    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 39 | SALE1030 | 2024-05-12 | CUST1124    | PROD73     | STORE0   | 1        | ₹ 328.51   | 20%      | COO          | ₹262.81      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 40 | SALE1031 | 2025-03-03 | CUST1274    | PROD44     | STORE4   | 1        | ₹ 34.51    | 4%       | COO          | ₹33.13       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 41 | SALE1032 | 2023-11-20 | CUST1105    | PROD19     | STORE11  | 3        | ₹ 663.73   | 6%       | COO          | ₹1,871.72    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 42 | SALE1033 | 2024-07-06 | CUST1246    | PROD33     | STORE9   | 1        | ₹ 409.41   | 13%      | PayPal       | ₹356.19      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 43 | SALE1034 | 2025-03-11 | CUST1003    | PROD78     | STORE3   | 3        | ₹ 461.03   | 11%      | PayPal       | ₹1,290.06    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

|    | A           | B                  | C   | D      | E                  | F              | G       | H             | I | J | K |
|----|-------------|--------------------|-----|--------|--------------------|----------------|---------|---------------|---|---|---|
| 1  | Customer_ID | Name               | Age | Gender | City               | State          | Country | Loyalty_Level |   |   |   |
| 2  | CUST1000    | Sheena Bonilla     | 55  | Female | Donaldson          | North Carolina | USA     | Unknown       |   |   |   |
| 3  | CUST1001    | Eugene Nelson      | 26  | Female | Phillipsmouth      | North Dakota   | USA     | Platinum      |   |   |   |
| 4  | CUST1002    | Dr. Benjamin Price | 29  | Female | Kylebury           | Alabama        | USA     | Platinum      |   |   |   |
| 5  | CUST1003    | Jeremy Harris      | 59  | Female | Thomastown         | Ohio           | USA     | Bronze        |   |   |   |
| 6  | CUST1004    | Caroline Collins   | 64  | Female | Nicholashire       | California     | USA     | Silver        |   |   |   |
| 7  | CUST1005    | Christine Vaughn   | 24  | Male   | East Amandamouth   | Oregon         | USA     | Bronze        |   |   |   |
| 8  | CUST1006    | Amber Jackson      | 29  | Female | Deborahborough     | South Dakota   | USA     | Platinum      |   |   |   |
| 9  | CUST1007    | Daniel Boyd        | 29  | Female | West Pamelaufurt   | West Virginia  | USA     | Bronze        |   |   |   |
| 10 | CUST1008    | Melissa Howard     | 53  | Female | Maynardton         | New Hampshire  | USA     | Gold          |   |   |   |
| 11 | CUST1009    | Ms. Maria Leach    | 64  | Male   | Melissashire       | Pennsylvania   | USA     | Platinum      |   |   |   |
| 12 | CUST1010    | Dominic Page       | 53  | Male   | Port Michaelburgh  | Pennsylvania   | USA     | Bronze        |   |   |   |
| 13 | CUST1011    | Brian Lopez        | 28  | Female | East Krista        | New York       | USA     | Gold          |   |   |   |
| 14 | CUST1012    | Kenneth Lambert    | 55  | Female | North Kathryn      | Kansas         | USA     | Platinum      |   |   |   |
| 15 | CUST1013    | Nathan Hernandez   | 32  | Female | New Becky          | Montana        | USA     | Silver        |   |   |   |
| 16 | CUST1014    | Ashley Cohen       | 53  | Female | Lake Kristamouth   | Nebraska       | USA     | Silver        |   |   |   |
| 17 | CUST1015    | Laura Alvarez      | 49  | Female | Lawsonhaven        | Washington     | USA     | Platinum      |   |   |   |
| 18 | CUST1016    | John Scott         | 24  | Female | South Christopher  | Pennsylvania   | USA     | Bronze        |   |   |   |
| 19 | CUST1017    | Adam Cruz          | 60  | Female | Brandonville       | Georgia        | USA     | Platinum      |   |   |   |
| 20 | CUST1018    | Rhonda Moss        | 43  | Male   | South William      | Kansas         | USA     | Unknown       |   |   |   |
| 21 | CUST1019    | Michael Torres     | 46  | Female | Gonzalezfort       | New Jersey     | USA     | Platinum      |   |   |   |
| 22 | CUST1020    | Bridget Allen      | 59  | Male   | Lake Shaneburgh    | Delaware       | USA     | Silver        |   |   |   |
| 23 | CUST1021    | Jeff Martinez      | 36  | Female | Smithburgh         | Oklahoma       | USA     | Gold          |   |   |   |
| 24 | CUST1022    | Paul Fox           | 33  | Female | Michaelton         | Hawaii         | USA     | Silver        |   |   |   |
| 25 | CUST1023    | Jonathan Steele    | 52  | Female | Molinalaby         | Oregon         | USA     | Silver        |   |   |   |
| 26 | CUST1024    | Heather Collins    | 40  | Male   | Rogerschesster     | Idaho          | USA     | Platinum      |   |   |   |
| 27 | CUST1025    | Diana Taylor       | 59  | Female | Vasqueztown        | California     | USA     | Bronze        |   |   |   |
| 28 | CUST1026    | Amber Carlson      | 18  | Female | East Michellemouth | Missouri       | USA     | Gold          |   |   |   |
| 29 | CUST1027    | Leslie Johnson     | 31  | Male   | Mollyshire         | Utah           | USA     | Unknown       |   |   |   |
| 30 | CUST1028    | Joshua Ryan        | 27  | Female | New David          | New Jersey     | USA     | Platinum      |   |   |   |
| 31 | CUST1029    | David Arnold       | 54  | Female | East Jennifer      | California     | USA     | Silver        |   |   |   |
| 32 | CUST1030    | Jessica Acevedo    | 64  | Female | Ambyburg           | Pennsylvania   | USA     | Bronze        |   |   |   |
| 33 | CUST1031    | Sarah Vance        | 41  | Male   | Johnsonville       | Oregon         | USA     | Silver        |   |   |   |
| 34 | CUST1032    | Mike Jennings      | 48  | Female | Fostermouth        | Arizona        | USA     | Gold          |   |   |   |
| 35 | CUST1033    | Robert Jones Dds   | 25  | Male   | Tristanton         | Oklahoma       | USA     | Platinum      |   |   |   |
| 36 | CUST1034    | Christopher Olson  | 27  | Female | West Richard       | Maryland       | USA     | Platinum      |   |   |   |
| 37 | CUST1035    | Rachel Myers       | 52  | Male   | Robertberg         | North Dakota   | USA     | Silver        |   |   |   |
| 38 | CUST1036    | Thomas Phillips    | 20  | Female | Lulsford           | Georgia        | USA     | Gold          |   |   |   |
| 39 | CUST1037    | Christopher Carter | 27  | Female | Brendastad         | South Dakota   | USA     | Unknown       |   |   |   |
| 40 | CUST1038    | Deanna Simpson     | 35  | Female | Brownport          | North Dakota   | USA     | Bronze        |   |   |   |
| 41 | CUST1039    | Tonya Reynolds     | 55  | Female | Darleneton         | Michigan       | USA     | Gold          |   |   |   |
| 42 | CUST1040    | Eric Vega          | 50  | Male   | Williamsonland     | Kansas         | USA     | Gold          |   |   |   |
| 43 | CUST1041    | James Gibbs        | 21  | Male   | Mooneyshire        | Alaska         | USA     | Platinum      |   |   |   |
| 44 | CUST1042    | William Martin     | 19  | Male   | Ryanburgh          | Connecticut    | USA     | Platinum      |   |   |   |
| 45 | CUST1043    | Joseph Snow        | 45  | Male   | South Michaelfort  | Vermont        | USA     | Gold          |   |   |   |

|    | A        | B        | C          | D          | E        | F          | G            | H            | I                                                                       | J | K       |
|----|----------|----------|------------|------------|----------|------------|--------------|--------------|-------------------------------------------------------------------------|---|---------|
| 1  | Sales_ID | Cust_id  | Order_date | Product_id | Quantity | Unit_Price | Payment_Type | Total_Amount |                                                                         |   |         |
| 2  | SALE999  | CUST1421 | 13-01-2024 | PROD81     | 4        | ₹ 22.51    | PayPal       | ₹ 65.73      |                                                                         |   |         |
| 3  | SALE998  | CUST1407 | 18-08-2024 | PROD34     | 3        | ₹ 336.17   | COD          | ₹ 968.17     |                                                                         |   |         |
| 4  | SALE997  | CUST1190 | 28-05-2025 | PROD27     | 3        | ₹ 530.17   | PayPal       | ₹ 1,304.22   |                                                                         |   |         |
| 5  | SALE996  | CUST1388 | 14-10-2024 | PROD48     | 4        | ₹ 826.55   | Credit Card  | ₹ 2,380.46   | here I get my required database from the slaes_fact table using vlookup |   |         |
| 6  | SALE995  | CUST1394 | 22-03-2024 | PROD24     | 3        | ₹ 480.11   | PayPal       | ₹ 1,281.89   |                                                                         |   |         |
| 7  | SALE994  | CUST1329 | 26-09-2025 | PROD99     | 2        | ₹ 314.40   | PayPal       | ₹ 584.78     |                                                                         |   |         |
| 8  | SALE993  | CUST1363 | 18-02-2024 | PROD19     | 2        | ₹ 957.54   | COD          | ₹ 1,723.57   |                                                                         |   |         |
| 9  | SALE992  | CUST1000 | 10-10-2024 | PROD40     | 3        | ₹ 490.02   | Credit Card  | ₹ 1,087.84   |                                                                         |   |         |
| 10 | SALE991  | CUST1319 | 20-05-2024 | PROD22     | 1        | ₹ 904.92   | Credit Card  | ₹ 787.28     | Descriptive Statistics for total_amount                                 |   |         |
| 11 | SALE990  | CUST1077 | 06-08-2025 | PROD20     | 3        | ₹ 955.82   | COD          | ₹ 2,408.67   |                                                                         |   |         |
| 12 | SALE99   | CUST1325 | 06-07-2024 | PROD54     | 3        | ₹ 541.99   | COD          | ₹ 1,235.74   | Mean                                                                    |   | 1085.19 |
| 13 | SALE989  | CUST1443 | 06-02-2025 | PROD88     | 1        | ₹ 66.14    | COD          | ₹ 56.88      | Standard Error                                                          |   | 18.3153 |
| 14 | SALE988  | CUST1004 | 13-07-2024 | PROD47     | 2        | ₹ 274.09   | PayPal       | ₹ 476.92     | Median                                                                  |   | 866.05  |
| 15 | SALE987  | CUST1026 | 13-10-2024 | PROD25     | 1        | ₹ 531.99   | PayPal       | ₹ 441.55     | Mode                                                                    |   | #N/A    |
| 16 | SALE986  | CUST1483 | 17-05-2024 | PROD51     | 3        | ₹ 461.02   | PayPal       | ₹ 981.97     | Standard Deviation                                                      |   | 818.882 |
| 17 | SALE985  | CUST1055 | 11-04-2025 | PROD1      | 4        | ₹ 880.26   | COD          | ₹ 3,133.73   | Sample Variance                                                         |   | 670568  |
| 18 | SALE984  | CUST1491 | 14-09-2025 | PROD29     | 4        | ₹ 465.42   | COD          | ₹ 1,377.64   | Kurtosis                                                                |   | 0.06741 |
| 19 | SALE983  | CUST1088 | 20-05-2024 | PROD66     | 2        | ₹ 853.93   | PayPal       | ₹ 1,690.78   | Skewness                                                                |   | 0.88874 |
| 20 | SALE982  | CUST1165 | 29-10-2025 | PROD31     | 4        | ₹ 538.73   | Credit Card  | ₹ 2,090.27   | Range                                                                   |   | 3889.78 |
| 21 | SALE981  | CUST1445 | 08-07-2024 | PROD85     | 2        | ₹ 943.93   | PayPal       | ₹ 1,472.53   | Minimum                                                                 |   | 18.459  |
| 22 | SALE980  | CUST1232 | 12-12-2023 | PROD66     | 4        | ₹ 649.52   | Credit Card  | ₹ 2,312.29   | Maximum                                                                 |   | 3908.24 |
| 23 | SALE98   | CUST1131 | 15-10-2024 | PROD56     | 4        | ₹ 476.28   | Credit Card  | ₹ 1,581.25   | Sum                                                                     |   | 2169301 |
| 24 | SALE979  | CUST1328 | 24-04-2025 | PROD21     | 3        | ₹ 614.63   | PayPal       | ₹ 1,733.26   | Count                                                                   |   | 1999    |
| 25 | SALE978  | CUST1298 | 21-03-2025 | PROD19     | 3        | ₹ 677.46   | Credit Card  | ₹ 1,930.76   |                                                                         |   |         |
| 26 | SALE977  | CUST1099 | 19-07-2025 | PROD31     | 4        | ₹ 540.69   | COD          | ₹ 1,968.11   |                                                                         |   |         |
| 27 | SALE976  | CUST1249 | 05-02-2025 | PROD19     | 2        | ₹ 466.42   | PayPal       | ₹ 820.90     |                                                                         |   |         |
| 28 | SALE975  | CUST1303 | 21-12-2023 | PROD16     | 2        | ₹ 61.42    | PayPal       | ₹ 113.01     |                                                                         |   |         |
| 29 | SALE974  | CUST1444 | 10-01-2025 | PROD10     | 4        | ₹ 599.85   | Credit Card  | ₹ 1,799.55   |                                                                         |   |         |
| 30 | SALE973  | CUST1402 | 20-01-2024 | PROD54     | 2        | ₹ 528.30   | Credit Card  | ₹ 803.02     |                                                                         |   |         |
| 31 | SALE972  | CUST1406 | 24-03-2025 | PROD3      | 1        | ₹ 402.47   | PayPal       | ₹ 326.00     |                                                                         |   |         |

|    | A          | B                           | C           | D                            | E        | F                               | G | H            | I |
|----|------------|-----------------------------|-------------|------------------------------|----------|---------------------------------|---|--------------|---|
| 1  | Product_id | product_name                | category    | brand                        | cost     |                                 |   |              |   |
| 2  | PROD0      | Brother Laser Printer       | Electronics | Hunt Inc                     | ₹ 112.42 |                                 |   |              |   |
| 3  | PROD1      | Men's Graphic T-Shirt       | Clothing    | Terry-Moore                  | ₹ 342.56 |                                 |   |              |   |
| 4  | PROD2      | Women's Casual Hoodie       | Clothing    | Sims Inc                     | ₹ 21.34  |                                 |   |              |   |
| 5  | PROD3      | Men's Running Jacket        | Clothing    | Pineda Ltd                   | ₹ 218.48 | Descriptive Statistics for Cost |   |              |   |
| 6  | PROD4      | Women's Yoga Pants          | Clothing    | Gonzales-Davis               | ₹ 326.06 | Mean                            |   | 240.5169     |   |
| 7  | PROD5      | Men's Leather Belt          | Clothing    | Baldwin, Garza and Gonzalez  | ₹ 43.29  | Standard Error                  |   | 13.2477637   |   |
| 8  | PROD6      | Herbal Face Moisturizer     | Beauty      | Davidson PLC                 | ₹ 17.31  | Median                          |   | 228.53       |   |
| 9  | PROD7      | LED Table Lamp              | Home        | Harrell LLC                  | ₹ 218.22 | Mode                            |   | 254.04       |   |
| 10 | PROD8      | Wireless Bluetooth Earbuds  | Electronics | Perez-Johnston               | ₹ 228.53 | Standard Deviation              |   | 132.477637   |   |
| 11 | PROD9      | Adjustable Dumbbell Set     | Sports      | Hill PLC                     | ₹ 183.04 | Sample Variance                 |   | 17550.32429  |   |
| 12 | PROD10     | Ceramic Kitchen Bowl Set    | Home        | Hamilton Group               | ₹ 342.56 | Kurtosis                        |   | -0.954646171 |   |
| 13 | PROD11     | HD Smart TV Stick           | Electronics | Gibson, Hobbs and Weaver     | ₹ 449.40 | Skewness                        |   | 0.112129881  |   |
| 14 | PROD12     | Hair Strengthening Shampoo  | Beauty      | Rice LLC                     | ₹ 115.82 | Range                           |   | 467.31       |   |
| 15 | PROD13     | USB-C Fast Charger          | Electronics | Dawson-Perkins               | ₹ 242.23 | Minimum                         |   | 17.31        |   |
| 16 | PROD14     | Decorative Wall Mirror      | Home        | Luna, Anderson and Bowman    | ₹ 48.01  | Maximum                         |   | 484.62       |   |
| 17 | PROD15     | 4K Home Projector           | Electronics | Thomas LLC                   | ₹ 465.34 | Sum                             |   | 24051.69     |   |
| 18 | PROD16     | Beard Grooming Kit          | Beauty      | Erickson-Green               | ₹ 337.42 | Count                           |   | 100          |   |
| 19 | PROD17     | Soccer Training Cone Set    | Sports      | Ford, Murphy and Wilson      | ₹ 144.53 |                                 |   |              |   |
| 20 | PROD18     | Women's Denim Jeans         | Clothing    | Guerra-Hendrix               | ₹ 254.04 |                                 |   |              |   |
| 21 | PROD19     | Knee Support Brace          | Sports      | May-Reese                    | ₹ 21.34  |                                 |   |              |   |
| 22 | PROD20     | Outdoor Sports Water Bottle | Sports      | Soto-Ellis                   | ₹ 131.51 |                                 |   |              |   |
| 23 | PROD21     | Multi-Purpose Storage Box   | Home        | Gardner-Montes               | ₹ 123.46 |                                 |   |              |   |
| 24 | PROD22     | Carbon Fiber Tennis Racket  | Sports      | Flores, Pineda and Rodriguez | ₹ 484.62 |                                 |   |              |   |
| 25 | PROD23     | Men's Formal Shirt          | Clothing    | Moon, Mora and Santos        | ₹ 254.04 |                                 |   |              |   |
| 26 | PROD24     | Foldable Camping Chair      | Sports      | Cook Ltd                     | ₹ 166.06 |                                 |   |              |   |
| 27 | PROD25     | Women's Scarf Set           | Clothing    | Brown and Sons               | ₹ 174.02 |                                 |   |              |   |
| 28 | PROD26     | Wireless Bluetooth Speaker  | Electronics | Reed-Schmidt                 | ₹ 322.41 |                                 |   |              |   |
| 29 | PROD27     | Decorative Floor Vase       | Home        | Khan, Jones and Paul         | ₹ 254.04 |                                 |   |              |   |
| 30 | PROD28     | Sports Stopwatch Timer      | Sports      | Burns-Hansen                 | ₹ 38.76  |                                 |   |              |   |
| 31 | PROD29     | Fabric Sofa Throw Blanket   | Home        | Hoffman Group                | ₹ 218.48 |                                 |   |              |   |

|    | A        | B                               | C      | D                  | E          | F | G |
|----|----------|---------------------------------|--------|--------------------|------------|---|---|
| 1  | store_id | store_name                      | region | city               | Store_Type |   |   |
| 2  | STORE0   | Rivera-Warner                   | East   | Washingtonfurt     | Flagship   |   |   |
| 3  | STORE1   | Martinez, Adkins and Johnson    | North  | Danielmouth        | Flagship   |   |   |
| 4  | STORE2   | Black-Horn                      | South  | North Nancyview    | Outlet     |   |   |
| 5  | STORE3   | Johnson-Wright                  | North  | Hoodberg           | Outlet     |   |   |
| 6  | STORE4   | Alexander Group                 | South  | Port Sharon        | Online     |   |   |
| 7  | STORE5   | Gray Ltd                        | North  | New Michaelchester | Outlet     |   |   |
| 8  | STORE6   | Benson-Elliott                  | East   | Port Josephberg    | Online     |   |   |
| 9  | STORE7   | Martinez-Mullins                | South  | Williamsfurt       | Online     |   |   |
| 10 | STORE8   | Bean, Jones and Park            | West   | Thompsonborough    | Online     |   |   |
| 11 | STORE9   | Moore, Padilla and Butler       | South  | East Johnstad      | Flagship   |   |   |
| 12 | STORE10  | Hill, Wagner and Ballard        | East   | East Davidville    | Online     |   |   |
| 13 | STORE11  | Aguirre-Keller                  | South  | Lake Michael       | Online     |   |   |
| 14 | STORE12  | Thomas, Bridges and Harris      | North  | Sabrinatown        | Online     |   |   |
| 15 | STORE13  | Gardner, Humphrey and Contreras | East   | Johnsonville       | Flagship   |   |   |
| 16 | STORE14  | Mcdonald Inc                    | South  | Christopherland    | Outlet     |   |   |
| 17 | STORE15  | Shelton PLC                     | North  | East Jenniferstad  | Flagship   |   |   |
| 18 | STORE16  | Frederick-Wright                | West   | New Jimmymouth     | Flagship   |   |   |
| 19 | STORE17  | Holloway Ltd                    | North  | Bensonhaven        | Flagship   |   |   |
| 20 | STORE18  | Guerrero-Schmitt                | East   | East Stevechester  | Flagship   |   |   |
| 21 | STORE19  | Mann-Austin                     | East   | Vasquezborough     | Online     |   |   |
| 22 |          |                                 |        |                    |            |   |   |
| 23 |          |                                 |        |                    |            |   |   |
| 24 |          |                                 |        |                    |            |   |   |
| 25 |          |                                 |        |                    |            |   |   |
| 26 |          |                                 |        |                    |            |   |   |
| 27 |          |                                 |        |                    |            |   |   |
| 28 |          |                                 |        |                    |            |   |   |
| 29 |          |                                 |        |                    |            |   |   |
| 30 |          |                                 |        |                    |            |   |   |

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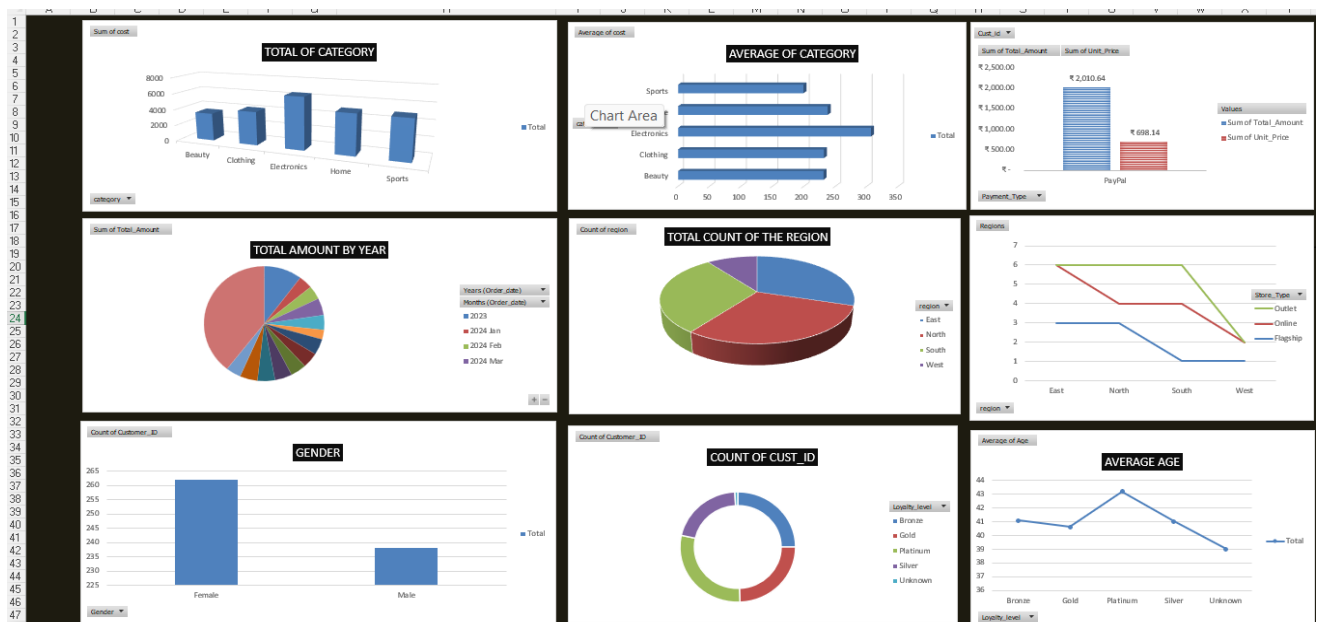
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## Visualization and Insights Arrived with screenshot of visualization Report/Dashboard screenshot:

- Descriptive analysis:
  - What is the total sales?
  - Which region performance is best?
  - Which year had the highest sales?
  - Which product category generated the highest sales?
- Diagnostic Analysis:
  - Why did the top category generate higher sales?
  - Why is the profit goes higher and lower for certain products?
- Predictive Analysis:
  - Based on the sales trend, what couldn't be the next months sales?
  - Which category is likely to perform well in the future ?
- Prescriptive Analysis:
  - Which category should the company focus on?
  - Should the company increase or reduce discounts?

### Dashboard Image:



### **Final 4 type of insights as storytelling:**

The analysis indicates that the business generated approximately 2.17 million in total sales, with 2024 contributing around 1.09 million, reflecting strong overall sales performance. The customer analysis identifies a total of 500 customers, comprising 262 female and 238 male customers, indicating a relatively balanced customer distribution. The Platinum loyalty segment represents the largest customer group, with 145 customers. Product cost analysis shows an average cost of 240.52 and a median of 228.53, with values ranging from 17.31 to 484.62, highlighting considerable variation in product costs. Overall, the analysis provides meaningful insights into sales performance, customer demographics, loyalty segments, and product cost patterns, supporting more informed business decision-making.

### **Conclusion:**

The Excel-based analysis provides a comprehensive view of the business's sales performance, customer characteristics, product costs, and store-level performance. Through data cleaning, transformation, formulas, Pivot Tables, descriptive statistics, and dashboard visualizations, the dataset was converted into meaningful business information. The analysis highlights strong overall sales performance, a balanced customer base, and clear differences in customer loyalty and product costs. These findings can support data-driven decision-making, customer-focused strategies, product planning, and improved business performance.